

Vicente Arroyos

arroyv@uw.edu
185 E Stevens Way NE
Seattle, WA 98195

Education

PhD Candidate

2020 - 2025

University of Washington
Paul G. Allen School of Computer Science & Engineering
3.9/4.0

Bachelor of Science

2016 - 2020

University of Washington
Department Electrical & Computer Engineering
3.4/4.0

Publications

1. Kyle Johnson, **Vicente Arroyos** et al. Designing a Flybar for Stabilizing Millimeter-scale Flying Robots. The IEEE Robotics and Automation Letters (in submission).
2. **Vicente Arroyos**, Kyle Johnson, et al. Battery-free shape changing origami for sensor deployment. Science Robotics (in submission).
3. **Vicente Arroyos**, Maria LK Viitaniemi, Nicholas Keehn, Vaidehi Oruganti, Winston Saunders, Karin Strauss, Vikram Iyer, and Bichlien H. Nguyen. "A Tale of Two Mice: Sustainable Electronics Design and Prototyping." In *CHI Conference on Human Factors in Computing Systems Extended Abstracts*, pp. 1-10. 2022.
4. Yasuda, Hiromi, Kyle Johnson, **Vicente Arroyos**, Koshiro Yamaguchi, Jordan R. Raney, and Jinkyu Yang. "Leaf-like origami with bistability for self-adaptive grasping motions." *Soft Robotics* (2022).

Presentations

1. **Arroyos, V.** *A Tale of Two Mice: Sustainable Electronics Design and Prototyping*. Poster presentation at CHI Conference on Human Factors in Computing Systems, New Orleans, LA
2. **Arroyos, V.**, Johnson, K. (2019). *Leafout Origami: Dynamic Fabrications and Applications*. Oral Presentation at the UCLA Ronald E. McNair 2019 Conference, Los Angeles, CA.
3. **Arroyos, V.** (2018) *Computer Science Educational Outreach Through the Summer Institute and CyPhyTown*. Oral Presentation at the Science Undergraduate Laboratory Internships (SULI) Program Summer Research Symposium, Richland, WA.

Entrepreneurship

UW, NSF I-Corps, CoMotion. "Designing a Flybar for Stabilizing Millimeter-scale Flying Robots"	Oct 2022
AVELA (A Vision for Electronic Literacy & Access), Nonprofit Co-Founder	Paperwork Pending 2022
MIT, NSF I-Corps, Spark Program. "AVELA"	Apr 2022

Media Coverage

1. [These Tiny Sensors Ride the Wind Like Dandelion Seeds](#)" CNET, May 20, 2022
2. "Q&A: Making Earth-friendly electronics" UW NEWS, Apr. 21, 2022

3. “A&A’s Australia Summer Program Helps Forge Future Directions With LSAMP”, William E. Boeing
Department of Aeronautics and Astronautics News, Dec. 2018

Research Experience

Iyer Lab, Sept. 2021 - Present

Graduate Researcher

Principal Investigator: Dr. Vikram Iyer

- Design and fabrication techniques for millimeter sized Leafout origami structures capable of jumping or gliding.
- Explore Leafout structure magnetic actuation for controlled jumping, locomotion, and state transformations
- Develop computing and sensor systems onto mm and cm scale structures taking advantage of their properties.
- Craft miniaturized mobile vehicle platforms for sensing and networking.
- Code Biodegradable and environmentally conscious circuits.

Oak Ridge National Lab (ORNL),

June 2021 - Aug. 2021

GEM Fellowship intern, Neutron Sciences Directorate

Mentor: Rob Knudson

- Processed 8000+ curves each with 100000+ points to extract 11 features.
- Used extracted features to visualize differences between gamma curves and neutron curves.
- Presented results at Oak ridge End of summer Symposium and scored 95 / 100 on my presentation.

Networks and Mobile Systems Lab,

June 2020 - Sept.2020

National Science Foundation Research Experiences for Undergraduates program intern

Principal Investigator: Dr. Shyam Gollakota

- Design and fabrication techniques for millimeter sized Leafout origami structure capable of jumping.
- Laser cut prototypes composed of carbon, PET, and Kapton to emulate dynamic simulation results.
- Explore Leafout structure magnetic actuation for controlled jumping and locomotion

Lab for Engineered Materials & Structures,

June 2019 - June 2020

Undergraduate Researcher

Principal Investigator: Dr. Jinkyu Yang

- Investigate design and fabrication techniques to create an energy efficient Leafout origami structure capable of jumping and grasping. Conduct kinematic simulation tests.
- Laser cut prototypes composed of Bristol, PET, nylon,
- Explore Leafout structure actuation for controlled frequency-based jumping.

Patents

Designing a Flybar for Stabilizing Millimeter-scale Flying Robots ROI Pending 2022 *Kyle Christopher Wynn Johnson, Vicente Arroyos, Raul Villanueva, Adriana Shulz, Sawyer Fuller, Vikram Iyer*

Grants

National Science Foundation S-STEM: 22-527 | **\$1,499,906**
Community Outreach Director

Under Review Sep 2022

“AVELA: Culturally Responsive Academic Support Systems for Low Income and Underrepresented Minority Communities”

Work Experience

Seattle Mathematics Engineering Science Achievement (MESA), Nov. 2016 - Jan. 2019

Program Student Assistant

- Engaged a diverse group of 6-12 Seattle School students from the Mukilteo, Tukwila, and the Seattle school districts in STEM activities that connect mathematics and science fundamentals to the real world one on one and class demos.
- Assisted with the Seattle MESA’s annual MESA Day STEM competition promotion and event development

Pacific Northwest National Lab (PNNL), June 2018 - Aug. 2018

Department of Energy SULI Program Intern

- Prepared course materials and helped teach introductory and advanced Scratch (block-based coding program) and electrical circuits, at the WSU Tri-Cities and PNNL Computer Science Teacher Education Project Summer Institute –a week-long K-12 teacher training on computational thinking and concepts for the classroom.
- Continued use of Raspberry Pi units to develop cyber physical system called CyPhyTown, a core component of PNNL’s annual Pink Elephant unicorn competitive outreach event, to demonstrate and teach cyber-physical security in a real-world setting using low-cost electronics.

Teaching and Mentorship

A Vision for Electronic Literacy & Access (AVELA), Mar. 2019 - Present

(Co-founder) Financial Officer,

- Fundraising to grow the organization Manage Academic Service statements
- Manage Organization and student funding.
- Mentor student project developments.
- AVELA’s objective is to provide opportunities for minority and underrepresented students to pursue their interests in college and engineering by leading workshops, camps, and other forms of community outreach by representing the populations we aim to support
- We work with: The City of Seattle, Seattle Public Schools, The Urban League, WA MESA, & more

UW College of Engineering

Math Academy Tutor, June 2020 - July 2020

- Co-led a team of 4 students to create two activities in line with common core mathematics standards
- Presented activities weekly for 4 weeks to approximately 50 students
- Exposed student to python coding and graphing
- Exposed students to data collection and visualization

Engineering Peer Educator (EPE), Mar. 2019 - Dec. 2019

- Lead a class of 23 direct to college undeclared engineering students weekly for one hour.
- Support new students in their transition to the UW College of Engineering community.
- Create an engaging discussion section experience that contributes to Engineering Undeclared students’ achievement of the GEN ST 199 learning outcomes.

Math Academy Tutor, May 2019 - June 2019

- Co-led and planned a 3-hour Arduino activity for a group of 36 high school students coming to the University of Washington (UW) for Math Academy. Implemented Arduino Nanos and implemented a 6-

axis gyroscope and accelerometer to measure impact force in circuit drop activity for summer programs to increase student interest in engineering.

- Goal: exposure to engineering concepts and aspects of design, problem-solving, testing, analysis, force, acceleration, and programming circuits.

Early Engineering Institute (EEI) Tutor,

May 2019 - June 2019

- Co-led engineering activity for EEI summer sessions one and two to increase K-14 student interest in engineering, each activity was 2 hours in length
- Co-Developed an Arduino lesson for each group of ~60 middle school students at EEI so that students could apply physics concepts from the classroom to real world situations using coding/circuitry tools.

Louis Stokes Alliance for Minority Participation (LSAMP) in STEM,

Nov. 2018 – Jan. 2020

Ambassador

- Collaborated in the planning and facilitating of the 2019 PNW LSAMP Conference, hosting approximately 300 minority students in STEM from 5+ institutions in the Pacific Northwest.
- Participated in recruitment efforts, including tabling and co-leading a 3D print workshop for 10 students to increase LSAMP student interest.

Projects

Cyberworks Robotics Autonomous Wheelchair (industry Capstone Project),

Jan. 2020 - June 2020

Student

- Identification of navigational confounds in a complex large scale indoor/outdoor environment
- Review Cyberworks navigation stack documentation.
- Create graceful motion 30-page implementation manual and report.

ARM Pipelined CPU (Computer Architecture I Final Project),

Sept. 2020 - Dec. 2020

Student

- Used knowledge of Assembly language programming, Computer Arithmetic, Processor Data paths & Control, Memory hierarchy, and caches to Develop and design a Verilog implementation of a 64-bit ARM CPU with a 5 stage Pipelining.
- Simulated the digital CPU's 27 modules and ~1100 lines of code in ModelSim
- In assembly executed a sorting algorithm, a function returns the Nth number in the Fibonacci sequence

Happy Feet Embedded Systems Final Project),

June 2020

Student

- Designed a dancing robot (Happy Feet) created by configuring an LCD and servo registers on a Texas Instrument Board to communicate with each other to actuate different motions.
- utilized IAR's workbench and ~2000 lines of code for Texas Instruments TM4C123GH6PM board.

FPGA Invaders (Class Final Project),

May 2019

Student

- Imitated the popular game Space Invaders by utilizing an FPGA board, Keyboard and Speaker
- Designed, simulated, and built a digital system using System Verilog, Quartus, and ModelSim
- Move the tank left and right using the A and D keys respectfully. Shoot the aliens, score some points or the game is over.

Design Challenge (Industry Challenge),

May 2019

Student

- Proposed unique features for the next generation of SONOS speakers
- Developed prototype software for a smart speaker experience that delivers the right sound, at the right moment, into people's lives.

- Used a Python run API to send commands to a locally run webpage that controlled various functions on the SONOS speaker; like playing, pausing, and adjusting the speaker's volume.

Services

Gabriel E. Gallardo Research, Student Leadership & Advocacy Symposium	Moderator,	Apr. 2022
Gabriel E. Gallardo Research, Student Leadership & Advocacy Symposium	<i>Panelist,</i>	Apr. 2022
Electrical & Computer Engineering: Undergraduate orientation,	<i>Presenter,</i>	Sept. 2019
PNW LSAMP Leadership & Training Seminar,	<i>Panelist,</i>	Aug. 2019
Upward bound Workshop,	<i>Panelist,</i>	June. 2019
Future Student Visit Day,	<i>Tabling,</i>	June. 2019
UW Admitted Student Preview Day,	<i>Panelist,</i>	Apr. 2019
Seattle MESA,	<i>University Tour,</i>	Oct. 2018
LSAMP 3D Printing and Design,	<i>Presenter,</i>	Jan. 2019
UW's STEM Study Abroad in Australia,	<i>Presenter,</i>	Jan. 2018

Awards and Affiliations

ACM Student member,	2022 - 2023
WSGC Fellow,	2021 - 2022
GEM Fellow,	2020 - Present
Ronald E. McNair Postbaccalaureate Achievement Program Scholar,	2019 - 2020
Vincent Kerkof Endowed Scholarship,	2019
James M. and Virginia C.B. Webster Endowed Scholarship,	2019
Plan to Go Scholarship,	2018
National Action Council for Minorities in Engineering (NACME) Scholarship,	2018
UW Dean's List,	Dec. 2018
Washington State Opportunity Scholarship (WSOS) Scholar,	2016 - 2020
UW Brotherhood Initiative,	2016 - 2020
Office of Minority Affairs and Diversity (OMAD) UW LSAMP,	2016 - 2020

Other Education / Training

Aeronautics and Astronautics Design of Novel Materials and Structures
 Queensland University of Technology (QUT), Aug. 2018 - Sep 2018